Code No. 3512 /S

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) V-Semester (SUPPLEMENTARY) Examinations Jul/Aug-2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Why is Turing Test significant in AI?	2	1	1
b)	Define state space representation with an example.	2	3	2
c)	Identify the importance of inference mechanisms in AI.	2	2	3
d)	Define machine learning and list its main types.	2	2	4
e)	What is fuzzy logic? Mention one application.	2	2	5
f)	Differentiate informed search from uninformed search?	2	3	1
2. a)	Write and Explain about A* algorithm with an example.	6	2	1
b)	Given a sliding tile puzzle with three tiles and a blank (0 represents blank) with the following initial and final states. Assume the moves to be blank right, then left then up and then down in that order. Exhibit the first 4 nodes that will be expanded by DFS and BFS.	6	4	1

0	1
3	2

Initial State

1	2
0	3

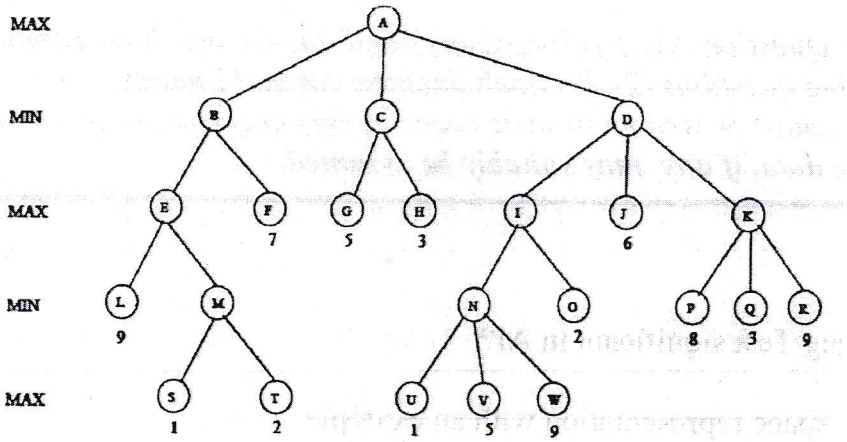
Goal State

3. a) Evaluate the game tree where the root is a maximizing node, and children are visited left to right. 6 4 2

a) Compute the minmax game value of each node using the standard minmax algorithm.

(b) What move will be selected by player one using minmax.

(c) List the nodes (leaves or interior nodes) that alpha-beta algorithmprunes, i.e., decides need not be



b) What are frames? Give a sample frame of a Computer department of a college. 6 3 2

4. a) Describe Fuzzy inference system with an example. 6 2 3

b) Explain the architecture of an Expert System with an example. 6 2 3

5. a) Explain entropy and information gain. Discuss their role in decision tree construction using a suitable example. 6 3 4

b) Demonstrate the structure and working of artificial neural networks using OR gate. 6 2 4

6. a) Describe the process of syntactic analysis in NLP. How do parsing techniques help in understanding sentence structure? 6 2 5

b) Explain the need for probabilistic models in Statistical Natural Language Processing (NLP). 6 2 5

7. a) Consider the following sentences: Translate these sentences into formulas in predicate logic. 6 3 2

i) John likes all kinds of food.

ii) Apples are food

iii) Chicken is food

iv) Anything anyone eat and isn't killed by is food.

v) Bill eats peanuts and is still alive.

vi. Sue eats everything Bill eats.

b) Discuss Sequence of steps taken in designing a fuzzy logic machine. 6 2 3

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